

Physics Department
Fisika Departement

Semester Test 1

Semestertoets 1

FSK116

14 March 2013 / 14 Maart 2014

Surname and initials

Van en voorletters

Student number

Studentenommer

Signature

Handtekening

Time

2 hours

Total

65 marks max

Tyd

2 ure

Totaal

65 punte maks

Internal Examiner:

Dr J M Nel

External Examiner

Mr R Q Odendaal

Interne Eksaminator

Eksterne Eksaminator

- Answer all questions in the space provided.
- Clearly show all calculations. Do not just write down the final answer.
- Clearly state all assumptions, rules and laws used.
- Where applicable draw the free-body diagrams.
- Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks.
- Programmable calculators may NOT be used.
- **NO** textbooks allowed.
- A cheat sheet may be used, but must adhere to the rules printed on it.
- All test and examination regulations of the University of Pretoria are applicable.

- *Beantwoord alle vrae in die spasie gelaat.*
- *Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.*
- *Stel duidelik alle aannames, reëls en wette wat u gebruik.*
- *Waar van toepassing, teken die vryliggaamsdiagram.*
- *Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.*
- *Programmeerbare sakrekenaars MAG NIE gebruik word nie.*
- ***GEEN** handboeke word toegelaat nie.*
- *'n Formuleblad mag gebruik word onderhewig aan die reëls daarop gedruk.*
- *Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

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Semester Test 1 **FSK 116** **Semestertoets 1**
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Venue

Lokaal

Question 1

- a) Calculate (to an accuracy of 2 significant figures) the number of breaths taken during an average human lifetime. You can assume that a typical human lifetime is 70 years and the average number of breaths that a person takes is 10 breaths per minute. Clearly show all your units and conversions which you use during the calculation.

Bereken (tot 'n akkuraatheid van 2 beduidende syfers) die aantal asemteue wat 'n gemiddelde mens in 'n leeftyd inasem. Neem aan dat 'n tipiese persoon 'n leeftyd het van 70 jaar, en 'n gemiddeld van 10 asemteue per minuut inasem. Wys duidelik al die eenhede en omskakelings wat jy in die berekening gebruik.

[4]

- b) A burette of which the cross-sectional area, A , is constant, contains water to a height $h = h_0$. When the stopcock is opened the volume of water flows out at a rate of $A \frac{dh}{dt}$. If the stopcock is opened slightly, the height of the water in the tube is given by the function $h = h_0 e^{-\lambda t}$ metres in which the time, t , is measured in seconds and λ is a positive **constant** with unit s^{-1} . Determine the rate (volume per unit time) at which the water flows from the burette and explain the negative sign of the answer.

*'n Buret met 'n konstante deursnee-oppervlakte, A , is gevul met water tot by 'n hoogte $h = h_0$. As die kraantjie oopgemaak word, is die volume water wat per tydseenheid uitvloei, $A \frac{dh}{dt}$. As die kraantjie baie klein oopgedraai word, word die hoogte van die water gegee deur die funksie $h = h_0 e^{-\lambda t}$ meter waar die tyd, t , in sekonde gemeet word en λ 'n positiewe **konstante** is met eenheid s^{-1} . Bepaal die uitvloeitempo (volume per tydseenheid) van die water en verklaar die negatiewe teken van die antwoord.*

[4]

- c) Calculate the area bounded by the t -axis and the graph of the function $v = 3 + 4t$, between $t = 1$ s and $t = 5$ s. The units of v are metres per second and that of t is seconds. What are the units of the quantity which the area represents?

Bereken die oppervlakte begrens tussen die t -as en die grafiek van die funksie $v = 3 + 4t$, tussen $t = 1$ s en $t = 5$ s. Die eenhede van v is meter per sekonde en dié van t is sekondes. Wat is die eenhede van die oppervlakte wat bereken is?

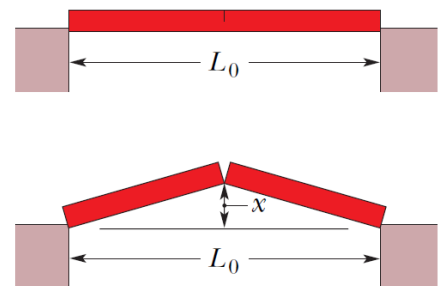
[4]

Question 2

- a) As a result of a temperature rise of ΔT , a bar with a crack at its centre buckles upward. If the fixed distance is L_0 and the coefficient of linear expansion of the bar is α :

'n Gekraakte staaf buig opwaarts in die middle as gevolg van 'n ΔT toename in die temperatuur. Indien die afstand L_0 vas is en die koëffisiënt van lineêre uitsetting is α :

- Find the rise x of the centre.
Bepaal die afstand x wat die middelpunt opwaarts beweeg.



[4]

- If the temperature rise is 32°C , the distance $L_0 = 3.77$ m and the coefficient of linear expansion of the bar is $25 \times 10^{-6}/^\circ\text{C}$, what is the value of x ?
Indien die temperatuurverandering 32°C is, die afstand $L_0 = 3.77$ m en die koëffisiënt van lineêre uitsetting is $25 \times 10^{-6}/^\circ\text{C}$, wat is die waarde van x ?

[2]

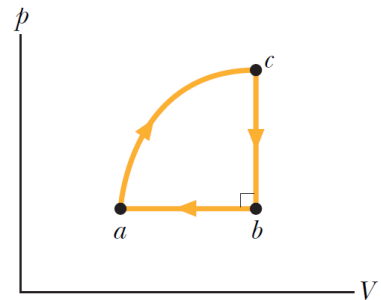
- b) The temperature of a 0.700 kg cube of ice is found to be -150°C . Energy is gradually transferred to the cube as heat while it is otherwise thermally isolated from its environment. The total heat transferred is 0.6993 MJ. Assume that the specific heat of ice, $c = 2220 \text{ J/kg}\cdot\text{K}$ is valid for the all temperatures from -150°C to 0°C . What is the final state of the H_2O and what is the final temperature? $L_f = 333 \text{ kJ/kg}$, $L_v = 2256 \text{ kJ/kg}$, $c_{\text{water}} = 4187 \text{ J/kg}\cdot\text{K}$

Die temperatuur van 'n 0.700 kg blok ys is -150°C . Energie in die vorm van warmte word gelydelik oorgedra aan die blok wat termies geïsoleer is van die omgewing. Die totale hoeveelheid warmte oorgedra is 0.6993 MJ. Veronderstel dat die spesifieke warmte van ys, $c = 2220 \text{ J/kg}\cdot\text{K}$, is geldig vir alle temperature vanaf -150°C tot 0°C . Wat is die finale toestand van die H_2O en wat is die finale temperatuur? $L_f = 333 \text{ kJ/kg}$, $L_v = 2256 \text{ kJ/kg}$, $c_{\text{water}} = 4187 \text{ J/kg}\cdot\text{K}$.

[7]

- c) The figure displays a closed cycle for a gas. For c to b , 40 J is transferred from the gas as heat. From b to a , 130 J is transferred from the gas as heat, and the magnitude of the work done on the gas is 80 J. From a to c , 400 J is transferred to the gas as heat. What is the work done by the gas from a to c ? Clearly show all your reasoning.

Die figuur toon 'n geslote siklus vir 'n gas aan. Van c tot b word 40 J oorgedra vanuit die gas as warmte. Van b tot a word 130 J oorgedra vanuit die gas as warmte, en die hoeveelheid arbeid gedoen op die gas is 80 J. Van a tot c word 400 J oorgedra na die gas as warmte. Hoeveel arbeid word gedoen deur die gas van a tot c ? Wys duidelik met redes hoe jy jou die antwoord bereken.

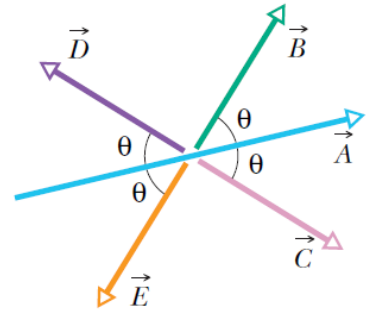


[5]

Question 3

- a) The figure shows vector $\vec{A}$, and the four vectors $\vec{B}$, $\vec{C}$, $\vec{D}$ and $\vec{E}$, that have the same magnitude but differ in orientation. Which of the four vectors $\vec{B}$, $\vec{C}$, $\vec{D}$ and $\vec{E}$ have the same dot product with $\vec{A}$?

Die figuur vertoon vektor $\vec{A}$, en vier ander vektore $\vec{B}$, $\vec{C}$, $\vec{D}$ en $\vec{E}$, wat dieselfde grootte maar verskillende oriëntasies het. Watter van die vier vektore $\vec{B}$, $\vec{C}$, $\vec{D}$ en $\vec{E}$ het dieselfde puntproduk met $\vec{A}$?

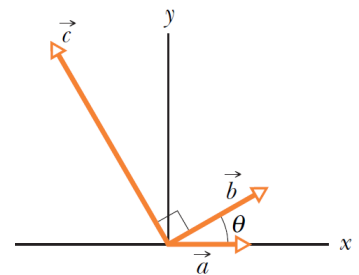


[4]

- b) The three vectors in the Figure have magnitudes $a = 3.00$ m, $b = 4.00$ m and $c = 10.0$ m. The angle $\theta = 30.0^\circ$.

Die drie vektore in die figuur het groottes $a = 3.00$ m, $b = 4.00$ m en $c = 10.0$ m. Die hoek $\theta = 30.0^\circ$.

- Write these vectors in unit vector notation.
Skryf die vektore in eenheidsvektor-notasie.



[6]

- Determine the magnitude and direction of the cross product of $\vec{b} \times \vec{c}$.
Bepaal die grootte en rigting van die kruisproduk $\vec{b} \times \vec{c}$.

[3]

Question 4

- a) When the acceleration is **constant**, the average and instantaneous acceleration are equal:

$$a = a_{avg} = \frac{v - v_0}{t - 0}, \text{ where } a \text{ is the acceleration, } v \text{ is the velocity at time } t \text{ and } v_0 \text{ is the initial}$$

velocity at $t = 0$ s. We can also write the average velocity as $v_{avg} = \frac{x - x_0}{t - 0}$, where x is the position

at time t and x_0 is the initial position at $t = 0$ s. Using these two equations (**not the integration method**) show that when a particle is accelerating in the x direction, its velocity is given by $v^2 = v_0^2 + 2a(x - x_0)$. Show all steps and state all reasoning clearly.

*Wanneer die versnelling **konstant** is, is die gemiddelde en oombliklike versnelling dieselfde:*

$$a = a_{gem} = \frac{v - v_0}{t - 0} \text{ waar } a \text{ die versnelling is, } v \text{ is die snelheid by tyd } t \text{ en } v_0 \text{ is die aanvanklike}$$

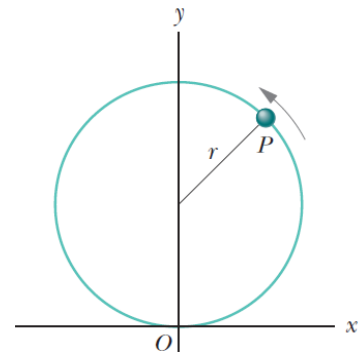
snelheid by } t = 0 \text{ s. Ons kan ook die gemiddelde snelheid skryf as } v_{gem} = \frac{x - x_0}{t - 0}, \text{ waar } x \text{ die}

*posisie by tyd } t \text{ is en } x_0 \text{ die aanvanklike posisie by } t = 0 \text{ s is. Deur gebruik te maak van dié twee vergelykings (**nie die integrasie-metode nie**), bewys dat die snelheid gegee word deur } v^2 = v_0^2 + 2a(x - x_0) \text{. Wys al die stappe en verduidelik jou aannames.}*

[9]

- b) A particle P travels with constant speed on a circle of radius $r = 3.00$ m and completes one revolution in 20.0 s. The particle passes through O at time $t = 0$ s. The following vectors must be given in terms of magnitude and their direction from the positive x -axis.

'n Deeltjie P beweeg met konstante spoed op 'n sirkel met radius $r = 3.00$ m en dit voltooi een omwenteling in 20.0 s. Die deeltjie gaan deur O wanneer $t = 0$ s. Die volgende vektore moet in terme van die grootte en rigting ten opsigte van die positiewe x -as gegee word.



- The position vector at $t = 5.00$ s.
Die posisie-vektor by $t = 5.00$ s.

[2]

- The position vector at $t = 10.0$ s.
Die posisie-vektor by $t = 10.0$ s.

[2]

- The displacement of the particle during the time $t = 5.00$ s and $t = 10.0$ s.
Die verplasing van die deeltjie in die tydsinterval $t = 5.00$ s en $t = 10.0$ s.

[2]

- The acceleration at $t = 5.00$ s.
Die versnelling by $t = 5.00$ s

[3]

- c) A cameraman on a pickup truck is traveling westward at 20 km/h while he records a cheetah that is moving westward 30 km/h faster than the truck. Suddenly, the cheetah stops, turns, and then runs at 45km/h eastward, as measured by a suddenly nervous crew member hiding behind a bush alongside the cheetah's path. The change in the animal's velocity takes 2.0 s.

'n Kameraman staan agter op 'n bakkie wat in 'n westelike rigting beweeg teen 20 km/h terwyl hy 'n jagluiperd opneem wat ook in 'n westelike rigting beweeg teen 30 km/h vinniger as die bakkie. Die jagluiperd stop skielik, draai, en hardloop dan in 'n oostelike rigting teen 45 km/h soos gemeet deur 'n baie senuweeagtige spanlid wat agter 'n bos wegkruip langs die jagluiperd se pad. Die verandering in die dier se snelheid neem 2.0 s.

- What are the magnitude and direction of the animal's acceleration according to the cameraman?
Wat is die grootte en rigting van die dier se versnelling volgens die kameraman?

[5]

- What are the magnitude and direction of the animal's acceleration according to the nervous crew member?
Wat is die grootte en rigting van die dier se versnelling volgens die senuweeagtige spanlid?

[2]

ENJOY THE WEEKEND / GENIET DIE NAWEEK
